Demand Response Management System (DRMS)
Product Requirements Document

1. Purpose

The Demand Response Management System (DRMS) enables utilities to reduce or shift electricity demand during periods when the grid is under stress or when demand approaches available supply capacity.

The system is primarily intended to:

- Prevent grid overload during peak demand conditions
- Reduce the need for load shedding or blackouts
- Lower the cost of electricity procurement during peak periods

The system achieves this by coordinating controlled reduction or shifting of electricity consumption across participating consumers and devices.

2. User Roles

Utility Operator – manages programs and events
Grid Manager – oversees grid performance
Consumer – participates in programs and events
Administrator – configures and manages system settings

3. Customer Segmentation

Customer segmentation enables the identification and grouping of consumers and devices for targeted participation in Demand Response (DR) events. It provides the foundation for selecting appropriate participants to achieve desired load reduction.

Segmentation is used in both:

- **Manual DR execution**, where operators select predefined groups
- **Automatic DR execution**, where the system dynamically identifies optimal participants

Types Of Segmentation:

Static Segments

The system shall allow operators to create and manage predefined segments based on:

- Device attributes (e.g., EV charger type, capacity)
- Electrical attributes (e.g., meter ID, transformer ID, feeder)
- Tariff or rate plan
- Geographic location
- Customer category (residential, commercial, industrial)

Operators shall be able to:

- Create, edit and delete segments (Only when not active in any program)
- Define filtering criteria
- View segment size and composition

Dynamic (Automatic) Segments

For automatic DR events, the system shall be capable of generating **temporary, event-specific segments** based on real-time conditions.

The system shall:

- Evaluate all eligible customers and devices
- Identify those capable of contributing to demand reduction
- Select an optimal subset to meet the required reduction target
- Create a temporary segment associated with the specific DR event

These segments shall not require manual configuration and shall exist onl y for the duration of the event.

Segmentation in Manual DR Events

When creating a DR event manually, the system shall allow the operator to:

- Select one or more predefined segments
- Apply filters to refine the target group
- View estimated metrics before execution, including:
    - Number of participants
    - Estimated load reduction potential
    - Historical participation trends (if available)

The system shall ensure that:

- Selected segments align with the program under which the event is created
- Only eligible and enrolled customers are included

Segmentation in Automatic DR Events

For automatic DR execution, the system shall:

- Automatically determine the required load reduction (The amount of the energy need to be curtailed to meet the high demand at that time)
- Evaluate customer eligibility based on:
    - Device controllability
    - Available flexible load
    - Historical responsiveness
    - User preferences and constraints
- Rank customers based on their suitability for participation
- Select a combination of customers sufficient to meet the target reduction, with an appropriate buffer to account for uncertainty

The system shall generate an event-specific segment for execution and track its performance.

Segment Insights and Estimation

For all segments, the system shall provide:

- Total number of customers/devices
- Estimated connected load
- Estimated controllable load
- Historical participation rates (if applicable)
- Expected response reliability (if available)

These insights shall assist operators in selecting appropriate segments and planning DR events.

Enhancements and Improvements over version 1:

- **BUG** = Only those segments should be visible while creating an event, that has been previously selected for creating a program

- No preview of customer count before saving a segment. When you set filters and hit Save, you have no idea if 5 customers match or 5,000 match

- Segments cannot be edited at all. If the user makes a typo in a filter, they must disable the segment and recreate it from scratch. So it would be better if there is option to edit a segment. This case is only eligible if the segment is not added in any event or program.

- Instead of Disabling segment it would be better if we would directly delete it.

- If a utility has 50+ segments, finding the right one without a search bar is painful. Add a search bar and make columns sortable.

- If a segment has zero matching customers either at creation or later due to data changes, the system should proactively inform the admin.

- Inside each segment's detail view, segment show a list of all DR events and programs that the segment is currently taking part in so that the admin can understand what all segments are remaining which haven't taken part in any event yet.

- While creating a segment you can see the no. of customers.

- Segment can only be created or deleted (only if it does not have any programs going live) but cannot be updated during live events .

- Segment analytics should also be there, like how much energy this segment has saved also how many events are linked to this segment.

- While creating a segment a calendar pops up, and then all the dates that have been entered as the starting date, and the ending date appears selected in the calendar, and admin can dis-select any of the dates.

4. Demand Response Programs

Demand Response (DR) Programs define structured, long-term initiatives under which demand response events are executed. A program establishes the scope and serves as a framework for organizing multiple DR events over a defined period.

Programs are created and managed manually by the utility and represent business-level configurations rather than operational actions.

The program module shall enable the utility to define and manage long-term DR initiatives and associate multiple DR events under a common structure. Multiple segments can be part of one program, and one segment can be part of multiple programs at the same time, given the DR events of one program don't clash with events from another program.

Program Configuration

The system shall allow operators to create and manage programs with the following attributes:

- Program name and description
- Start date and end date
- Target demand reduction (optional)
- Applicable customer segments
- Eligible device types (e.g., EV chargers, thermostats)

Additional features (Program)

1. Analyze the bx reports to create analytics on what programs are working out, what customers are opting for, what events are more successful, what type or area or user are participating, these analytics from bx should be shown on the demand response section in the Ems etc.

2. Auto enroll: There should be an option to auto enroll in similar events/ programs options. Based on the events which customers have participated actively

5. Demand Response Events

Demand Response (DR) Events are time-bound operational actions executed to reduce or shift electricity demand in response to grid conditions.

An event represents the actual execution of demand response, during which selected customers and devices are requested or controlled to modify their consumption.

Each event is associated with a DR program and targets a specific set of customers, either through predefined segments or dynamically selected participants.

Functional Objectives

The event module shall enable the system to:

- Execute demand reduction or load shifting during defined time windows
- Target appropriate customers and devices based on program and segmentation
- Support both manual and automatic event creation
- Ensure reliable and measurable load reduction
- Adapt to real-time conditions during event execution

Event Types

The system shall support the following types of DR events:

Scheduled Events

- Pre-planned events created in advance
- Typically used for predictable peak periods

Automatic Events

- Triggered automatically based on system conditions such as:
    - Forecasted peak demand
    - Real-time load thresholds
    - Grid constraints
- Participant selection and execution are handled by the system

Event Creation (Manual)

The system shall allow operators to create DR events with the following inputs:

Event Details

- Event name
- Associated program
- Start time and end time
- Duration
- Desired load reduction (kW / MW)
- Selection of one or more predefined segments
- Optional filtering within selected segments

- Type of control action to be applied, such as:
    - Reducing EV charging rate
    - Adjusting thermostat setpoint

- Limiting device operation

Additional features (Events)

- Cloning an event and making an event recurring between specific dates and time.
- Recurring event scheduling like right now events expire one time, what if we can schedule it for a week.
- Events status filter should be there (like active, upcoming etc) , event filters should be there in historical events.
- **Automatic Event Creation**
  Automatic event and program creation based on the forecasted stats by our AI model. Instead of automatic event creation, there can be suggestions for the type of event or program that can be created for the to handle the predicted high demand.
- **Recommendation or Suggestions on future programs or event creation or even smart segment creation**
  Suggestions or recommendation on programs and event creation based on the forecasted stats that also contains an expected success rate (confidence score 82%) {means 82% chances of this event gets successful in achieving its target} , so that the admin can simply accept the suggestions and programs or events are automatically created .
  The admin even gets suggestions or recommendation on the segment creation

6. Controlling and setting load reduction of appliances

Sachin sir will tell.

7. Customer Enrollment & Participation

Customers need to be part of a Demand Response (DR) program to participate in events. Enrollment defines how much control the system has over their devices and how they will participate during events.

Enrollment Options

When a program is created, it shall have the following enrolment options:

Opt-In Enrollment

- Customers choose to join the program manually
- They agree to allow higher control over their devices
- The system can reduce their load by a significant amount during events
- Customers receive higher incentives for participation

Pre-Consented Enrollment

- Customers are automatically included in programs based on prior consent
- No manual action is required for each program or event
- The system applies limited control on their devices
- Load reduction is kept low (e.g., around 5%)
- Incentives are lower compared to opt-in participation

Participation in Events

- For **manual DR events**, customers participate based on their enrollment type:
    - Opt-in users → higher load reduction
    - Pre-consented users → limited load reduction
- For **automatic DR events**:
    - The system primarily uses pre-consented users
    - Load reduction is applied in small amounts (e.g., ~5%)
    - This ensures minimal disruption and faster response

Participation Tracking

The system shall:

- Track which customers participated in each event
- Record the level of load reduction applied
- Maintain history of participation for future use

8. Calculation of Energy Reduction

The system shall measure how much electricity consumption was reduced during a Demand Response (DR) event. This is used to check if the event was successful and to calculate incentives.

Baseline Calculation

The system shall estimate how much energy a customer would have consumed if the DR event had not occurred.

Baseline shall be calculated using:

- Historical consumption data of the customer
- Consumption during similar time periods (same hour of day) on previous days
- Adjustments for recent usage patterns (if required)

The system may use:

- Average consumption over the last few similar days
- Weighted average giving more importance to recent days

Actual Consumption During Event

The system shall measure actual energy consumption during the event using:

- Smart meter data
- Device-level data (if available, e.g., EV chargers, thermostats)

Consumption shall be recorded for the exact duration of the event.

Load Reduction Calculation

The system shall:

- Compare baseline consumption with actual consumption during the event
- Calculate the difference as the load reduced

If actual consumption is higher than baseline, the load reduction shall be considered zero.

Reporting

The system shall provide:

- Total load reduction for each event
- Customer-wise contribution

- Event-level performance summary

Usage

The calculated load reduction shall be used for:

- Evaluating event success
- Calculating customer incentives

9. Incentive & Settlement

The system shall calculate incentives for customers based on the amount of load reduction achieved during Demand Response (DR) events.

Incentive Calculation

The system shall:

- Calculate incentives based on energy reduced (kWh)
- Apply program-defined incentive rates
- Support different incentive levels based on participation type:
    - Opt-in users → higher incentives
    - Pre-consented users → lower incentives

Settlement

The system shall:

- Generate customer-wise incentive summaries
- Provide reports for integration with billing systems
- Ensure incentives are calculated only for verified load reduction

Constraints

- No incentive shall be given if no actual reduction is achieved
- Incentives shall be based on validated data from M&V

Additional features (Incentives)

- Dynamic Rewards: If there is high demand for energy, then we have a dynamic way to give high rewards for that specific event.

- Estimate Earning Section (monthly / quarterly): We can let users know the amount of earnings he can save/make by considering their site details , ev type (already there in app in my ev) , charger type etc. And giving an estimate to user how much he can save on monthly or quarterly basis if he complies readily complies with the dr programs and events .
- Discover Earnings Section: We can let users know the amount of energy points / credit points / real earnings he has got or saved daily, monthly, quarterly.
- Streak Maintenance: More rebates or incentives will be provided if a user maintains a streak (suppose he maintains of streak of opt ins from suppose 20 days so he will get more incentives), if he breaks the streak he will be reverted back to the base incentive.

10. Forecasting & Prediction

The system shall predict future electricity demand to identify potential peak periods where Demand Response (DR) may be required.

The system shall:

- Estimate expected electricity demand for upcoming time periods (e.g., next few hours or next day)
- Use historical consumption data and recent trends for prediction

Peak Detection

The system shall:

- Identify time periods where demand is expected to be high
- Flag potential peak conditions where DR events may be needed

Forecasting output shall be used to assist operators in planning manual DR events and suggesting automatic DR events.

12. Analytics & Reporting

The system shall provide insights into Demand Response (DR) performance at program, event, and customer levels.

Event Analytics

The system shall provide:

- Total load reduction achieved

- Actual vs target reduction
- Participation rate
- Event-wise performance summary

Program Analytics

The system shall provide:

- Number of events conducted
- Total reduction achieved under the program
- Overall participation trends

Customer-Level Insights

The system shall:

- Show individual customer contribution
- Track participation history
- Display incentives earned

Reporting

The system shall:

- Generate summary reports for events and programs
- Allow data export for further analysis or billing

13. Communication & Notifications

The system shall communicate Demand Response (DR) events to customers and provide real-time visibility into their participation through notifications and in-app interfaces.

Event Notifications

The system shall notify customers about:

- Upcoming DR events (for opt-in, if applicable)
- Start of an event
- End of an event

Notifications may be delivered via:

- Mobile app (push notifications)
- In-app messages (toast)

Customer Widget / In-App Display

The system shall provide a widget or in-app section that displays:

- Status of current live event (if any)
- Event timing and duration
- Participation status
- Basic information such as:
    - "Load reduction active"
    - "Event scheduled"

For upcoming events, the widget shall allow:

- Opt-in (if required)
- Viewing event details

Preferences

The system shall allow customers to:

- Enable or disable notifications (where applicable)
- View event-related information within the app

Other features:

- **Leaderboard** – We can add this (Ranking for each user will be given) based on that extra rebates can be added, like top 10 people will get better rebates
  Also there can be a comparison between neighbors. There will be 2 types of leaderboard event based and program based.

- Export feature should be there for extracting stats info in the pdf form etc

- Target and achievement graph should be there for the future understanding (more visual and graphical representation)

- **Graph to Monitor Grid Conditions**
  Graph that includes real-time grid usage + forecasting the future grid usage based on historical records (load and peak) and also taking into account the future scenarios.

- **Emergency opt Outs**
  A user gets a maximum of N monthly/quarterly /yearly/all-time opt outs from emergency automatic dr events in which he is auto enrolled .

- **Friction in opts out**
  While a user opts out an additional step
  a) An extra pop up suggesting an emotional message “ You will help power 10 power homes today” or “You will loose 20$ today If you opt out”.